

ECO387C - Fall 2024

Problem Set 4 Solutions

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Question 1 - Manual value function iteration (deterministic)

1. The state variable for this problem is the current period capital stock, or k . The control variables are consumption and the next period's capital, c and k' respectively.

2.

$$v(k) = \max_{\{0 \leq k' \leq k^\alpha\}} \{\log(k^\alpha - k') + \beta v(k')\}$$

3. The table below outlines the summary of the second iteration, assuming that the initial guess was $v_0(k) = 0$

$k \backslash k'$	0.04	0.08	0.12	0.16	0.20
0.04	-1.72	-1.71	-1.77	-1.88	-2.04
0.08	-1.49	-1.45	-1.48	-1.55	-1.64
0.12	-1.36	-1.31	-1.32	-1.37	-1.44
0.16	-1.27	-1.21	-1.21	-1.25	-1.31
0.20	-1.20	-1.13	-1.13	-1.16	-1.20

Applying the max operator for each k across the rows results in the following approximation of the value and policy function after the second iteration:

k	$v_2(k)$	$g_2(k)$
0.04	-1.7097	0.08
0.08	-1.4530	0.08
0.12	-1.3081	0.08
0.16	-1.2072	0.08
0.20	-1.1279	0.12

Question 2 - Manual value function iteration 2 (stochastic)

1. The state variables are y_t and a_t . The control variables are c_t and a_{t+1}
2. The Bellman equation is

$$v(a, y) = \max_{a' \geq a^{\min}} \left\{ \log \left(y + a - \frac{a'}{1+r} \right) + \beta \mathbb{E} [v(a', y')] \right\}$$

3. $v_2(a, y)$ is

$a \backslash y$	5	6
-1	2.61	2.80
0	2.80	2.95
1	2.95	3.09
2	3.09	3.21
3	3.21	3.31

The optimal policies associated with $v_2(a, y)$ are

Saving (a')

$a \backslash y$	5	6
-1	-1	-1
0	-1	-1
1	-1	-1
2	-1	-1
3	-1	0

Consumption (c)

$a \backslash y$	5	6
-1	4.95	5.95
0	5.95	6.95
1	6.95	7.95
2	7.95	8.95
3	8.95	9.00